Fiscal Year:	FY 2011	**Task Last Updated:**	FY 08/02/2011
PI Name:	Chen, Chung-Lung Ph.D.		
Project Title:	Heat Transfer on Multiscale Textures with Electrowetting Control		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	cchen@teledyne.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	805-373-4181
Organization Name:	Teledyne Scientific Company		
PI Address 1:	Thermal and Flow Physics		
PI Address 2:	1049 Camino Dos Rios		
PI Web Page:			
City:	Thousand Oaks	**State:**	CA
Zip Code:	91360-2362	**Congressional District:**	24
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	07/01/2009	**End Date:**	11/30/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	3
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
Contact Email:	david.f.chao@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: PI change for FY2012 necessitated change in end date to 11/30/2011 (previously 6/30/2012). See Song-Yong Park (PI) and project titled "Heat Transfer on Multiscale Textures with Electrowetting Control--(PI=Park)" for FY2012 report [Ed., 5/16/2012]		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Cheng, Jiangtao (Teledyne Scientific Company) Park, Sungyong (Teledyne Scientific)		
Grant/Contract No.:	NNC09CA36C		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Editor's Note 5/16/2012: PI change for FY2012 necessitated change in end date to 11/30/2011 (previously 6/30/2012). See Song-Yong Park (PI) and project titled "Heat Transfer on Multiscale Textures with Electrowetting Control--(PI=Park)" for FY2012 report. Modern electronic systems used in space must be reliable and efficient with thermal management unaffected by outer space environments. Heat pipes and vapor chambers based on phase-change heat transfer mechanisms have proven to be an efficient approach to dissipate heat in many space or ground-based thermal-energy systems. However, a layer of capillary wick structure fully covering the chamber walls, e.g., condenser and evaporator surfaces, creates additional thermal resistance, because the effective thermal conductivity is much lower in saturated wicks. This thermal barrier can generate an unacceptable temperature rise, especially in high heat-flux applications. To overcome this issue, we propose to develop an innovative wickless and electrowetting-controlled vapor chamber with bioinspired two-tier textures on the chamber wall. These engineered superhydrophobic / superhydrophilic multiscale textures, formed by growing carbon nanotubes (CNTs) on top of micropillars on a silicon chip, mimic the structure of lotus leaves; they can promote dropwise condensation at the condenser site, and enhance film evaporation at the evaporator site. The condensate droplets will be dislodged from the condenser site and circulated to the heat source via electrowetting transport. Droplet transport by electrowetting without any mechanical moving parts has great potential in thermal management due to its programmable and reconfigurable characteristics, prompt response, and low power consumption. Most importantly, given a two-dimensional array of control electrodes, condensate can be transported over the low-friction hydrophobic surface to a hot spot without the significant pressure loss that is associated with fine pore wicks. The transported droplets will be captured by a few finger-like extensions from the heat source, which are hydrophilized by a coating of UV&Ozone-treated carbon nanotubes. This hydrophilic coating will force droplets to cling to the hot spot surface, changing heat transfer into the more efficient form of thin film evaporation. In this program, we aim to a) investigate wetting characteristics and phase-change heat transfer on multiscale textures; b) quantify electrowetting transport of condensates and transient heat transfer on the textured surface; c) develop numerical models of heat transfer on the textures. Reduced gravity flight tests will be performed to validate our analysis and ground tests. This innovative design from this program will enable a new generation of high-performance space integrated systems to operate at high-power densities without problems from temperature gradients, increased weight, or added complexity. Low-pressure-loss droplet transport on multiscale textures, combined with adaptable, low-power electrowetting, has the potential to significantly improve vapor chamber efficiency, increase its robustness and lifetime, and reduces the mass of space flight systems needed to support future human space exploration activities. The application of nanothermal materials and adaptive liquid transport technology based on electrowetting will lead to a major breakthrough in the thermal management field. The success of the proposed effort can considerably enhance thermal management in the future space explorations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	One of the major applications of phase-change heat transfer at low-gravity conditions is the design and maintenance of robust and efficient thermal control systems for future space exploration and satellite applications. Heat pipes and vapor chambers based on phase-change heat transfer mechanisms have proven to be good candidates to dissipate heat in many space thermal-energy systems. However, a layer of capillary wick structure fully covering the heat pipe and vapor chamber inner walls creates additional thermal resistance because the effective thermal conductivity is much lower in saturated wicks. This thermal barrier can generate an unacceptable temperature rise, especially in high heat-flux and harsh-environment applications. To address this issue, Teledyne Scientific & Imaging Company (TS&I) proposes to develop an innovative wickless and electrowetting-controlled vapor chamber with bioinspired two-tier textures on the chamber wall. This innovative design will enable a new generation of high-performance space integrated systems to operate at high-power densities without problems from temperature gradients, increased weight, or added complexity. Low-pressure-loss droplet transport on multiscale textures, combined with adaptable, low-power electrowetting, has the potential to significantly improve vapor chamber efficiency, increase its robustness and lifetime, and reduce the mass of space flight systems needed to support future human space exploration activities. The application of nanothermal materials and adaptive liquid transport technology based on electrowetting will lead to a major breakthrough in the thermal management field. The success of the proposed effort can considerably enhance thermal management in the future space explorations.
	Recent advances in high performance microelectronics technologies have resulted in tremendous improvements in functionality, power consumption, and energy efficiency. However many of the new electronic instruments operate at higher power densities and require high heat flux thermal management. The flexibility and stability of electrowetting-on-dielectric (EWOD) in combination with its cost effectiveness and portability make it an ideal platform to address a wide range of problems in chip cooling. During the first year of this program, we have successfully developed an EWOD-based microfluidic technique for active and adaptive thermal management of on-chip hot spots. A two-dimensional array of control electrodes was patterned on the chip surface for EWOD operations. By applying DC or AC voltages with appropriate sequence and timing to the electrode units, we were able to transport microdroplets of tens of microL along a programmable path. Without the need of external pumps and valves, the droplets were precisely delivered to the cooling targets. With EWOD-based microdroplet actuations, the hot spot surface was cooled down by more than 40ºC within 2 seconds. Furthermore, we demonstrated high heat flux (7.6 W/cm2) cooling on a hot spot and the EWOD-induced internal circulation within the droplets led to a time-averaged Nusselt number of ~45. However, under even higher levels of heat flux, the chip surface temperature can reach the Leidenfrost point, generally in the range of 150-180ºC, and an insulating vapor layer between the hot spot surface and the droplet base will form due to the Leidenfrost effect. Thus formed vapor layer keeps the droplet from evaporation/boiling rapidly and significantly lowers the cooling efficiency. In the worst case, due to the low friction of the vapor layer under the droplet, the convective vapor stream can easily drive the droplet away from the cooling target. Any deviation from the hot spot can lead to an instantaneous temperature increase and markedly reduce the cooling efficiency. Similarly, in our prior electrowetting cooling work using a sequence of water droplets, we also observed a rapid temperature increase after the removal of each droplet. Such a rapid temperature increase raised our concern of the chip device overheating. To overcome this issue, we implemented site-specific treatments on an EWOD device by selectively covering the hot

Task Progress:	spot surface (not the whole device surface) with a copper screen. The copper screen we used is SEFAR 100/.0045". The screen is hydrophilic and has a mesh size of 254 micrometers x 254 micrometers. Such a micro-porous screen can function as a wick structure that generates a strong capillary force to facilitate the delivered liquid droplets infiltrating into and permeating these micro-pores. The diameter of the copper wire is 0.0045" and one layer of the copper mesh is ~200 micrometers thick. Consequently a thin liquid film forms in the saturated copper screen which has a relatively low thermal resistance. As a result, heat transfer is in the more efficient form of filmwise evaporation at the evaporator sites. To maintain continued thin film evaporation, the EWOD-aided droplet splitting and merging are important to precisely confine the droplet volumes and hence the film thickness. We have successfully implemented droplet splitting in the parallel-plate configuration. Initially a droplet spans at least three control electrodes, i.e., occupies the entire middle electrode while overlapping a portion of the two neighboring electrodes. At first, we stretch the droplet in opposite directions by activating the left and right electrodes while grounding the middle electrode. As a result, the meniscus on the middle electrode starts to neck till the formed liquid finger breaks. Consequently EWOD-induced droplet elongation and necking gives rise to droplet splitting. Merging droplets is essential for mixing individual microdroplets to form a larger droplet, which is another effective EWOD operation for mass control in adaptive electrowetting cooling. Droplet merging is achieved by moving two microdroplets toward each other. For this purpose, EWOD actuations were simultaneously conducted on the electrode units under each droplet to drive the droplets together. Droplet merging can also be achieved by transporting a droplet to a stationary droplet. By EWOD-based droplet merging and splitting operations, we can accurately control the droplet volume, e.g., from 100 pL to 50 microL, to match the required film thickness depending on the real-time heat flux. Since EWOD-controlled droplet merging and splitting provides an efficient way to confine droplet volume, the EWOD-driven droplet prompt actuation with precise mass control indicates the significant potential to create an EWOD system for high heat flux cooling. For real applications, a closed loop of the EWOD cooling system is desired and has been implemented by us. The closed loop of our EWOD cooling system has an open-plate and parallel-plate combined configuration, i.e., droplet creation (splitting) from the reservoir occurs in the parallel-plate configuration while droplet imbibition into the copper screen takes place in the open-plate configuration. Without a cover plate over the hot spot, water vapor and heat can be easily convected away from the spot surface. In the closed loop system, comprehensive EWOD operations, such as successive droplet creation from the reservoir (condenser), prompt droplet transport and droplet permeation on the evaporator, were carried out simultaneously on the EWOD chip surface. During the second year of this program, heat dissipation has been significantly improved by implementing site-specific treatments on the chip surface. Based on EWOD technique with surface treatments, the superheat on a hot spot of 5mmx5mm was maintained well below 35ºC even when the heat flux reached as high as 80W/cm2. The closed loop of this novel thermal management system can potentially function as a wickless vapor chamber or heat pipe with enhanced heat dissipation capabilities.
Bibliography Type:	Description: (Last Updated: 05/11/2012)
Articles in Peer-reviewed Journals	Cheng J, Park S, Chen C-L. "Active Chip Cooling Using Electrowetting with Site Specific Treatment." International Journal of Heat and Mass Transfer. In review, July 2011. , Jul-2011
Articles in Peer-reviewed Journals	Cheng J-T, Chen C-L. "Active thermal management of on-chip hot spots using EWOD-driven droplet microfluidics." Experiments in Fluids 2010 Dec;49(6):1349-57. http://dx.doi.org/10.1007/s00348-010-0882-4 , Dec-2010
Papers from Meeting Proceedings	Cheng J, Chen C-L. "Active Chip Cooling Using Electrowetting." 42nd AIAA Thermophysics Conference, Honolulu, Hawaii, June 27-30, 2011. 42nd AIAA Thermophysics Conference, Honolulu, Hawaii, June 27-30, 2011. Paper AIAA 2011-3316. , Jun-2011